

CASE STUDY

The parts passed inspection. The defects shipped anyway.

They asked us to turn their paper inspection sheets into an app. The real problem: the data those sheets captured reached the quality team a shift late — so by the time a process drifted, suspect parts had already left the dock for the OEM.

Real Problem

“Build us an app to replace our paper inspection sheets. The clipboards are slowing the line.”

Scope

The clipboards weren’t the cost — the latency was. Quality data reached engineers a shift late, after suspect parts had already shipped to the OEM.

01. The Situation

A tier-2 automotive-components supplier in South Africa’s Eastern Cape fed just-in-time into the OEM assembly plants under IATF 16949 — running hundreds of line-side quality and machine checks every shift, all on paper.

Readings were keyed into the ERP and the customer scorecard a shift later. In automotive, a defect that reaches the OEM isn’t a return; it’s a PPM hit, a containment exercise, and a risk of stopping the customer’s line. The checks were being done correctly. The signal simply arrived too late to stop a bad batch leaving the building.

02. What We Found

Our diagnostic surfaced three compounding issues.

- 1. **Captured on paper, entered a shift later.** Quality data trailed the line by hours, so process drift was always caught after the fact.
- 2. **Containment stalled in email.** A quality hold could sit half a day while suspect stock kept moving toward dispatch.
- 3. **The same reading keyed into two systems.** The ERP and the OEM scorecard — transcription errors, wasted hours, and no live view of PPM trending.

75%

of new applications will be built on low-code by 2026 — up from under 25% in 2020. The tools are mainstream now; the edge is no longer building the app, but wiring its data to the decision it should trigger. Most rollouts stop at the form.

GARTNER LOW - CODE FORECAST

03. What We Did

We didn’t start with the app. We started with where the data needed to go.

Capture at the line

We rebuilt the checks as a Power Apps form on a tablet at the station —captured at source, validated against the control plan, with photos.

Contain in real time

A failed check auto-raises a quality hold in Power Automate, pages the engineer, and quarantines the batch— with the full audit trail IATF 16949 demands.

Connect + Copilot

Readings flow into the ERP and the customer scorecard — no re-keying — and Copilot answers plain-language questions like “what’s our PPM trend on Line 3?”

04. How We Rolled It Out

We didn’t try to boil the ocean. We piloted on a single line and proved the whole loop end to end — capture, containment, integration, query — then scaled across the lines. We kept operators and quality engineers in the build, so it fit how the floor actually runs. Within weeks, paper was of the pilot line and quality data was live.

05. What an Engagement Like This Delivers

Once quality data moved at the speed of the line, the gains followed — less manual work, faster containment, and fewer defects reaching the OEM.

85%

Less manual data entry

Re-keying into the ERP and scorecard eliminated.

40 min

Faster quality containment

Down from ~6 hours — quarantined before dispatch.

25%

Lower customer PPM

Fewer defective parts reaching the OEM.

The app wasn’t the answer. Catching the defect before the truck left was.

Bring us the problem

No proposal until we understand the problem.